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Department of Computer Science and Engineering

Project on

AI POWERED AUTONOMOUS COCONUT HARVESTING SYSTEM

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INTRODUCTION

India harvests nearly five billion coconuts annually, but coconut farming still depends heavily on manual tree climbing, which is risky, labour-intensive, and increasingly affected by skilled worker shortages.

Our AI-Powered Coconut Harvesting System combines drone imaging, AI-based tree and coconut detection, a tree-climbing harvesting robot, and a real-time web dashboard to automate safe and targeted harvesting. The system detects trees, classifies coconuts by ripeness, creates automated harvest tasks, and enables the robot to harvest only the required coconuts, improving safety, reducing labour dependency, and increasing efficiency in smart agriculture.

LITERATURE SURVEY

SL. NO	Author Name, Year of Publication	Title of the Paper, Journal of Publication	Methodology/Technique Implemented	Outcome (Advantages/Limitations)
1	Dr. SaravanaKumar R, Akshay Prasad Dubey, Santosh Mohan (2016)	*Autonomous Control and Implementation of Coconut Tree Climbing and Harvesting Robot*	Developed a climbing robot with DC motors and robotic cutting arm.	Reduced manual climbing effort but required manual supervision.
2	M. Kumar, G. Bej, H. S. Pandey (2023)	*Status and Scope of Automated Coconut Harvester in India: A Review*	Reviewed automated harvesters using AI and vision techniques.	Highlighted limitations and future scope of smart harvesting systems.
3	Hongcheng Zheng, Yuxing Fu, Zongbin Wang (2024)	*Design and Experiment of Coconut Picking Robot with Climbing*	Designed a climbing robot with chain-saw cutter and STM32 controller.	Faster harvesting but lacked autonomous AI vision support.
4	Samfford S. Cabaluna, Maria Fe P. Bahinting, Leah A. Alindayo (2024)	*Computer-Vision-Based Detection and Monitoring System for Mature Coconut Fruits with a Web Dashboard Visualization Platform*	Used YOLOv8 and computer vision for coconut maturity detection.	High detection accuracy but tested only in laboratory conditions.

LITERATURE SURVEY

SL. NO	Author Name, Year of Publication	Title of the Paper, Journal of Publication	Methodology/Technique Implemented	Outcome (Advantages/Limitations)
5	Mani A, Jothilingam A (2014)	*Design and Fabrication of Coconut Harvesting Robot: COCOBOT*	Developed a semi-autonomous robot with robotic arm and RF control.	Improved safety but lacked full automation.
6	V. Ramesh, Deepak Kumar (2025)	*AI and Sensor Based Smart Coconut Harvesting Machine*	Used AI algorithms and ultrasonic sensors for harvesting.	Reduced labor and improved harvesting efficiency.
7	R. Karthikeyan, S. Praveen (2023)	*AI Based Coconut Tree Climbing and Harvesting Robot*	Implemented AI-enabled climbing and cutting mechanisms.	Reduced human risk and increased harvesting speed.
8	Mohammed Afsal, N. Vivek (2024)	*Smart Coconut Harvesting System using IoT and Machine Learning*	Applied IoT sensors and machine learning for coconut detection.	Enabled smart monitoring and accurate harvesting.

LITERATURE SURVEY

SL. NO	Author Name, Year of Publication	Title of the Paper, Journal of Publication	Methodology/Technique Implemented	Outcome (Advantages/Limitations)
9	Priya Sharma, Arun Raj (2025)	*Deep Learning Based Coconut Detection for Automated Harvesting*	Used deep learning and CNN models for coconut detection.	Improved accuracy and reduced crop damage.
10	A. Lakshmi, P. Harish (2024)	*Autonomous Agricultural Harvesting Robot using Computer Vision*	Implemented autonomous navigation and object detection.	Increased productivity but affected by poor lighting conditions.
11	K. Suresh, M. Dinesh (2022)	*Robotic Arm for Coconut Harvesting in Agricultural Fields*	Developed robotic arm with wireless harvesting control.	Safer harvesting compared to manual climbing.
12	Arun Kumar, S. Naveen (2023)	*Machine Learning Based Smart Agriculture Harvesting System*	Used machine learning models and smart sensors for crop harvesting.	Improved automation and reduced harvesting time.

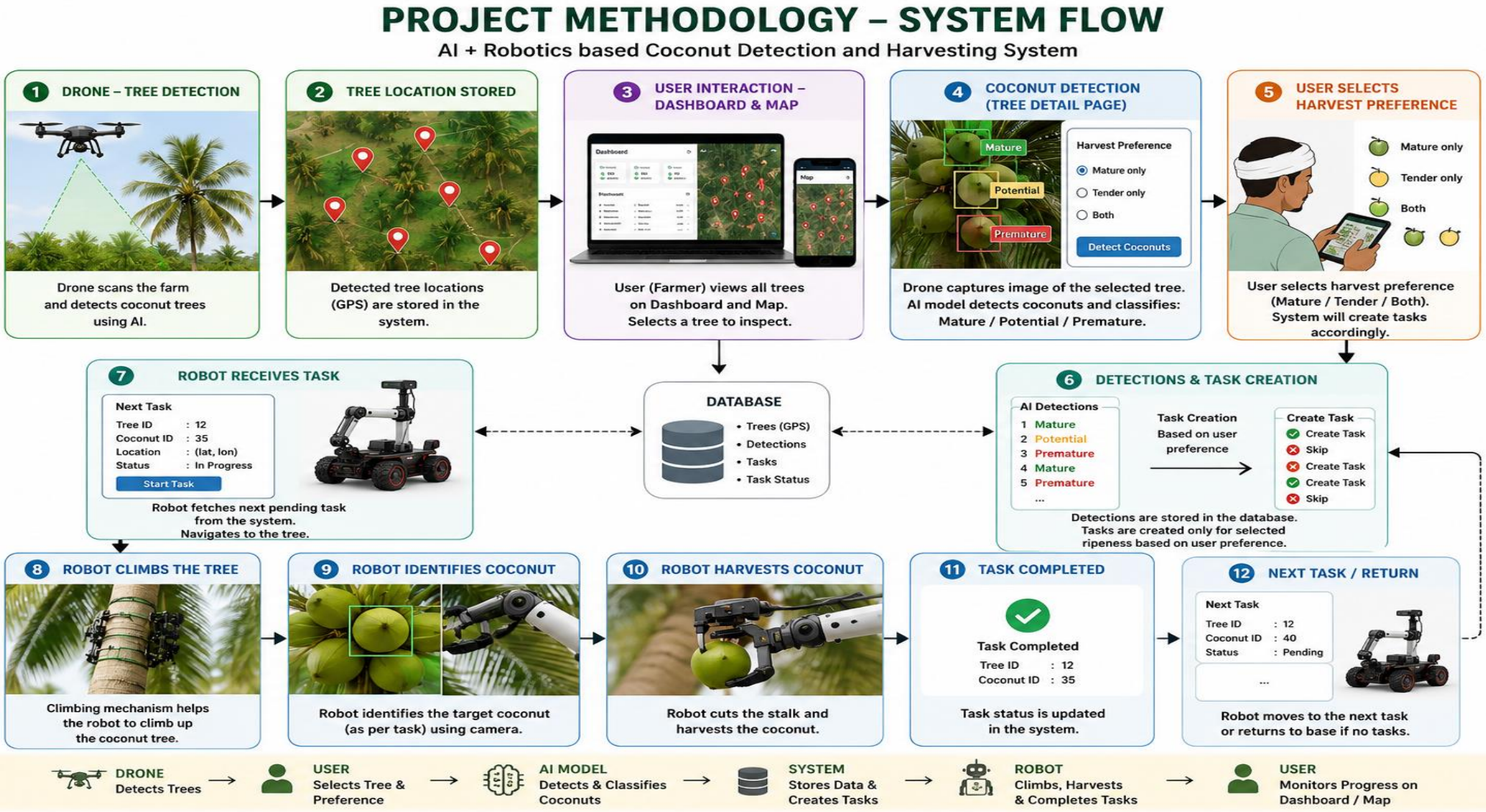
Problem Statement

Coconut harvesting in India continues to rely on manual tree climbing, a method that is physically dangerous, dependent on a shrinking pool of skilled workers, and increasingly uneconomical due to high labour costs. As plantation sizes grow across major producing states like Tamil Nadu, Kerala, and Karnataka, farmers have no efficient way to monitor tree readiness, identify mature versus tender coconuts, or coordinate harvesting at scale - leading to yield loss, missed harvests, and wasted labour. The absence of an intelligent, automated system results in poor farm productivity and growing safety risks, making it essential to develop a technology-driven alternative that can survey, detect, and harvest coconuts autonomously.

Objectives

- To automate the detection, GPS tagging, and location mapping of coconut trees using drone-captured aerial imagery and AI-based vision models.
- To classify coconuts by ripeness and generate farmer-controlled harvest task queues stored in a structured database.
- To enable a tree-climbing robot to autonomously execute harvesting tasks by communicating with a backend API in real time.
- To develop a web-based dashboard that provides farmers with live monitoring of plantation status, tree health, and harvest progress.

Methodology



Software and Hardware Requirements

Software Requirements

- **Backend-** Python, FastAPI, Uvicorn, PostgreSQL (viaDATABASE_URLenv), dotenv.
- **AI / Vision-** YOLOv8 (Ultralytics), OpenCV, Pillow for image processing and model inference.
- **Frontend-** Next.js 16, React 19, TypeScript, Tailwind CSS, React-Leaflet for map view.
- **Other-** Git for version control, Postman for API testing, VS Code as IDE.

Hardware Requirements

- **Drone-** GPS-enabled drone with a downward-facing camera for aerial plantation scanning.
- **Onboard Camera / Edge Device-** Camera module attached to the robot for close-up coconut ripeness detection.
- **Tree-Climbing Robot-** Motorised robotic unit capable of climbing coconut trees and cutting/plucking coconuts on command.
- **Processing Unit-** A laptop or server (minimum 8GB RAM, preferably with GPU) to run FastAPI backend and YOLO inference.

Advantages

- **Zero Manual Climbing** - Eliminates the need for workers to climb tall trees, significantly reducing the risk of fatal accidents and injuries.
- **AI-Driven Precision** - YOLOv8 models detect and classify coconuts by ripeness, ensuring only the right coconuts are harvested at the right time, reducing yield waste.
- **Scalable & Automated** - The task queue system allows one robot to work continuously across an entire plantation without human intervention, handling hundreds of trees efficiently.
- **Real-Time Farm Intelligence** - The live dashboard and GPS map give farmers instant visibility into every tree's status, coconut count, and harvest progress from any device.
- **Farmer-Controlled Preferences** - Farmers can choose to harvest mature coconuts, tender coconuts, or both giving full control over what gets harvested without changing the system.

Expected Outcome

- An automated drone scanning system capable of detecting and GPS-tagging coconut trees across any plantation without manual input.
- A trained YOLOv8 model that accurately classifies coconuts as mature or tender from camera images with consistent confidence.
- A smart task management system that generates and prioritises harvest tasks based on farmer preference and real-time detection data.
- A tree-climbing robot that operates independently by reading tasks from the API and reporting task completion back to the system.
- A responsive web dashboard with an interactive map, giving farmers complete visibility over every tree and harvest task on their farm.

References

- Wibowo, S. et al. (2016). *Autonomous Coconut Harvesting Robot.* — End-to-end robotic system with climbing and fastening mechanism for coconut plantations.scribd
- Rathod, A. et al. (2016). *Autonomous Control and Implementation of Coconut Tree Climbing and Harvesting Robot.* Procedia Computer Science, Elsevier.sciencedirect
- Amrita University (2015). *Prototype of a Coconut Harvesting Robot with Visual Feedback.* Advances in Intelligent Systems and Computing, Springer India.amrita
- IOSR Journal of Engineering (2025). *Coconut Harvest Mechanization Robot (CHMR) — Autonomous Device for Optimizing Coconut Harvesting.* Vol. 15, Issue 1.iosrjen
- Journal of Engineering and Applied Innovation (2023). *Status and Scope of Automated Coconut Harvester in India.* JEAI.journaljeai
- IJRMEE (2015). *Semi-Autonomous Coconut Harvesting Robot — Design and Development.* Vol. 2, Issue 10.ijrmee

References

- International Journal of Advanced Computer Science and Applications (2023). *Detection of Fruit using YOLOv8-based Single Stage Object Detection for Precision Agriculture.* IJACSA, Vol. 14, Issue 12.thesai
- Nature Scientific Reports (2024). *YOLOv8-CML: A Lightweight Target Detection Method for Intelligent Agriculture.* Nature, 2024.nature
- Frontiers in Plant Science (2025). *RLK-YOLOv8: Multi-Stage Detection of Fruits Using Improved YOLOv8.* Frontiers, Vol. 16.frontiersin
- CGIAR (2025). *From Sky to Soil: How Drones are Revolutionizing Crop Monitoring.* CGIAR Research, March 2025.cigar
- Journal of Agricultural and Crop Research (2025). *Optimizing Crop Monitoring Efficiency and Precision with Drone Technology.* ACRI, 2025.journalacri

References

- JOUAV (2023). *Agriculture Drones — Farming Drone for Crop Monitoring and Precision Farming.* jouav.com, October 2023.jouav

- Equinox Drones (2024). *Agriculture Drones for Precision Farming — UAV Aerial Imagery with Machine Learning.* equinoxsdrones.com, 2024.equinoxsdrones